A **biconnected graph** is a connected graph with the additional property that there is not a single vertex $v \in G$ such that removing v from G will disconnect the graph. An equivalent definition of biconnected graph can be given: G is biconnected if for any pair of vertices $u, v \in G$ there exist two disjoint paths between them; these two paths, therefore, can't have any edge in common, and neither any vertex except for u and v.

The proof of the first lemma is trivial: for a disconnected graph, since there are no edges between its connected components, it's sufficient to draw each component such that it won't overlap with the others.

As a consequence of the two lemmas, we can split any graph G into its biconnected components, and apply the planarity testing of choice to each of them separately.

15.2.3 A Naïve Algorithm for Planarity Testing

Since we have mentioned more than once that there is an (inefficient) algorithm based on Kuratowski's theorem that is fairly straightforward to implement, let's actually start from there. Listing 15.1 shows a template method that wraps any planarity testing algorithm, making sure we break down a graph into its connected (or, even better, biconnected) components, and running the testing on each of them.

In chapter 14 we have seen how we can find the connected components of a graph using DFS; finding biconnected